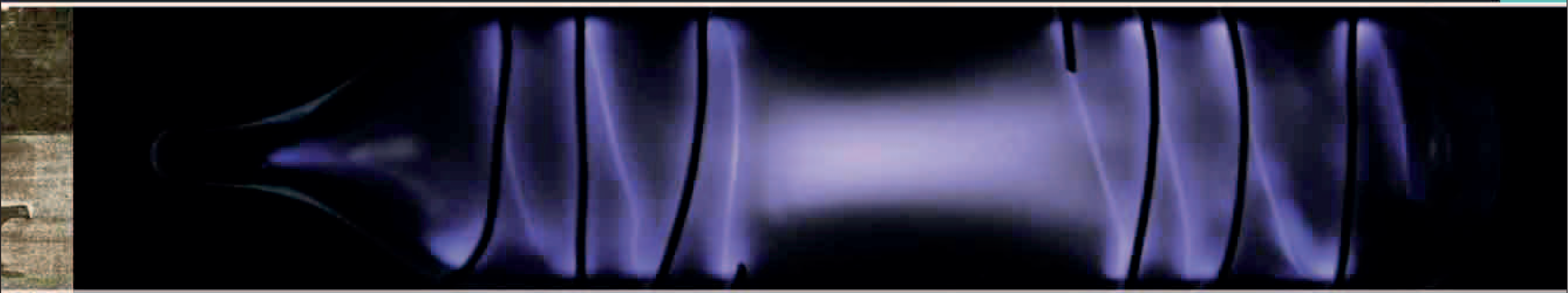

This lamp containing argon—first isolated in 1894—produces a violet discharge when placed in an electric field produced from a high voltage transformer to produce neon lighting.

RAMÓN Y CAJAL (1852–1934)

Spanish pathologist and histologist, Ramón y Cajal, was responsible for identifying a type of cell that controls the slow waves of contraction that move food along the intestine. He was also a neuroscientist and an expert in hypnotism, which he used to help his wife during labor. In 1906, he was awarded the Nobel Prize in recognition of his work on the nervous system.

and J.R. Benoît, Director of the International Bureau of Weights and Measures, decided to use wavelengths of light to redefine a standard of distance. They measured **the meter**—a prototype platinum–iridium bar of which was kept in Paris, France—in terms of the wavelength of the red light emitted by heated cadmium.

Edward Maunder (1851–1928), a British astronomer working at the Royal Greenwich Observatory in London, was investigating the historical records of sunspots in 1893 when he discovered that very few spots had been observed between 1645 and 1715. This interval, now known as the **Maunder Minimum**, coincided with the coldest part of the **Little Ice Age** (*c.*1500–1800), a time when Earth cooled considerably.

In 1894, British parasitologist **Patrick Manson** (1844–1922) developed the idea that **malaria is spread by mosquitoes**.

In November, Manson mentioned the hypothesis to British doctor **Ronald Ross** (1857–1932), who received the Nobel Prize in 1902 for working out the details of the process.

In 1784, British physicist Henry Cavendish had discovered that air contains a small proportion of a substance less reactive than nitrogen, but he was unable to isolate it. In August 1894, following a suggestion by British scientist Lord Rayleigh, British chemist **William Ramsay** (1852–1916) reported that he had isolated this gas, which he named **argon**. It was the first of the so-called **noble gases** to be isolated.

French engineer Édouard Branly (1844–1940) had developed an **early radio signal detector** at the beginning of the 1890s. In 1894, in lectures at the Royal Institution, London, British physicist **Oliver Lodge** (1851–1940) dubbed this the **coherer**. Lodge used this invention in his work, which became an important part of Italian physicist Guglielmo Marconi's system of wireless telegraphy.

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THE PERCENTAGE OF **ARGON** IN THE **ATMOSPHERE**

In 1894, Spanish histologist (histology is the study of tissues and cells) **Ramón y Cajal**, known as the father of modern neuroscience, theorized that **memories** do not involve growing new neurons (nerve cells), but making new connections between existing neurons. The connections between neurons came to be known as **synapses**.

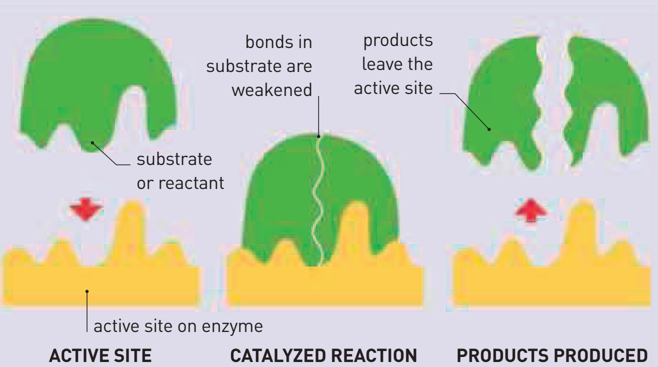
Also in 1894, British physiologist **Edward Sharpey-Schafer** (1850–1935) and English physician **George Oliver** (1841–1915) found that an extract from the adrenal gland caused a rise in blood pressure. This led them to identify the hormone **epineprine**.

German chemist **Emil Fischer** (1852–1919) in 1894 came up with the **lock and key theory** that explains how **enzymes target specific molecules** and function so efficiently.

> “THE **BRAIN** IS A **WORLD**… [WITH] A NUMBER OF **UNEXPLORED CONTINENTS** AND **STRETCHES** OF **UNKNOWN TERRITORY.**”
>
> **Ramón y Cajal, Spanish pathologist and histologist,** 1906

HOW ENZYMES WORK

Enzymes are proteins that act as catalysts to increase the rate of specific chemical reactions. They fold into complex shapes that allow smaller molecules to fit into them. The active site where these molecules fit may either encourage molecules to join together or split apart. However, the enzyme remains unchanged and can repeat the process indefinitely.



1894 Oliver Lodge improves the **coherer**, a radio signal detector

1894 Ramón y Cajal develops ideas about **brain function** and memory

1894 Edward Sharpey-Schafer and George Oliver identify the hormone **epinephrine**

1894 Emil Fischer explains how **enzymes** work

August 1894 William Ramsay reports isolating **argon** from the air

1894 Manchester Ship Canal is **completed**

1894 Lowell **Observatory** is founded in Arizona

1894 Guglielmo Marconi develops a **radio transmitter** that can ring a bell 33 feet (10 m) away

November 1894 Patrick Manson and Ronald Ross discuss the role of mosquitoes in spreading **malaria**